



102 genes linked to Autism

A study published in *Cell* revealed 102 genes linked to Autism, 37 of which were unlinked to the condition before. <https://digit.in/feb20-13>



Kill Switch for CRISPR

Anti-CRISPR proteins or 'kill switches' could make gene-editing safer and improve medical treatments. <https://digit.in/feb20-14>

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fforts to discover the genetic causes for several diseases such as cancer, schizophrenia and dementia are all

dependent on the thousands of studies relying upon the human gene list.

Upon receiving the DNA sequencing for a sick patient, usually the first question being asked is 'what genes are affected'. The assumption being that we know where the genes are located. Our knowledge regarding the Human gene catalogue is still far from complete despite the advancements in technology and medicine that we have seen in the last two decades.

The Human Genome Project (HGP), lasted from 1990 to 2003. This project was formulated to determine the DNA sequence and the "location of the estimated 100,000 human genes". It was believed that once the DNA sequence was determined, then it would be easier to determine where all the genes were located. With the rise in

competition amongst the human gene databases, despite the fact that the number of protein coding genes has gradually come together, the number of other gene types has exploded.

WHAT IS A GENE?

A gene is the basic physical and functional unit of heredity. Genes are made up of DNA. Certain genes act as instructions to make molecules called proteins. In the human genome, genes vary in size from a few hundred DNA bases to more than two million bases. Regions of the genome are transcribed into RNA which is then translated to create proteins. The Human Genome Project estimated that there are 20,000-25,000 human protein-coding genes. However, many genes were non-coding genes according to an original paper published in 2001 by the Human Genome Project which states, "thousands of human genes produce noncoding RNAs as the ultimate product". The paper reported

just 706 noncoding RNA genes. For this discussion we will take a look at this definition of Gene-

"A gene is any interval along the chromosomal DNA that is transcribed into a functional RNA molecule or that is transcribed into RNA and then translated into a functional protein."

This definition includes both non-coding RNA genes and protein-coding genes, while also grouping all the alternative splice variants at a single locus together, counting them as variants on the same gene. The non-functional remnants of true genes known as pseudogenes are excluded. However, this definition does raise the question as to what is meant by functional, and a truly comprehensive definition of the term gene might take too long to describe.

The human genome had begun with the assumption that the human genome contains 100,000 protein-coding genes, which was later revised in the 1990s reporting lower values between 50,000 and 100,000. The

COUNTING GENES

We don't really know how many genes we have, or even what exactly a gene is

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